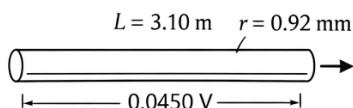


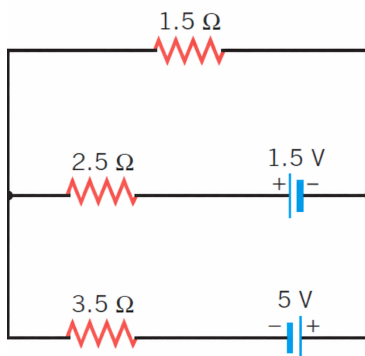
Final exam

Instructions: Solve five out of the following six problems.

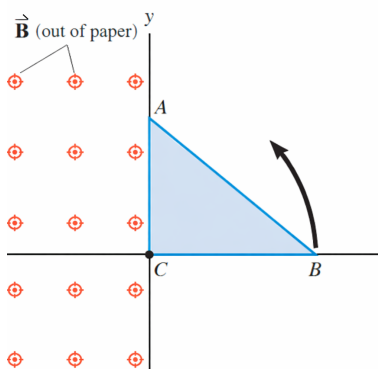
1. A cylindrical wire has a length of 3.10 m and a radius of 0.92 mm. When a potential difference of 0.0450 V is applied across its ends, a current of 1.10 A flows. Determine the resistivity of the wire.



2. Find the magnitude and direction of the current through the 2.5Ω resistor in the circuit shown below.



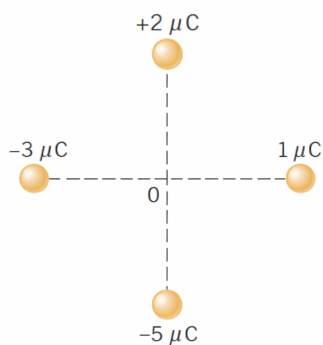
3. A rigid right triangle loop ABC made of aluminum rotates counter-clockwise about point C , which is located at the origin. A uniform magnetic field exists only to the left of the y -axis and is directed out of the page. As the triangle rotates, it crosses the positive y -axis. Determine whether the induced current is clockwise or counterclockwise at that moment. Justify your answer.



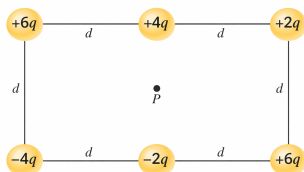
Please see next page!

Final exam

4. A transformer has 14 turns on the primary coil and 20 turns on the secondary coil. The secondary is connected to a resistor, and the peak voltage across the resistor is measured to be 54 V. What is the peak emf supplied by the generator connected to the primary coil?
5. Four point charges are placed on the coordinate axes, each at a distance of 7.5 cm from the origin. Calculate the magnitude of the net electric field at the origin.



6. Six point charges are arranged around a rectangle, where $q = 8.0\ \mu\text{C}$ and the spacing between adjacent charges is $d = 0.15\text{ m}$. Find the total electric potential at point P that is located at the center of the rectangle.



Good Luck!